

AP CSP | Lesson 28: Bias & Crowdsourcing

Homework | Multiple-choice and written practice

Name: _____ Class: _____ Date: _____

Time: 40 minutes | Total: 30 teacher marks (not an AP score) | Show working.

Original AP-style MCQs and classroom written practice; these are not official AP questions.

Q5: select exactly TWO. All cases and numbers are synthetic. Show calculations and distinguish model results from real evidence.

1. A fictional voluntary online survey gains many more responses from its existing users. What follows? [1 mark]

- A. Representation is guaranteed. B. More responses may preserve the same coverage gaps.
C. Non-users must be included. D. Every response is accurate.

Answer: _____

2. A fictional map ranks needs by raw report count. One person can resubmit the same report repeatedly. The main concern is: [1 mark]

- A. The list must be sorted alphabetically. B. All repeated observations are invalid.
C. The rule may reward repetition rather than need. D. The Internet cannot carry many reports.

Answer: _____

3. Which fictional activity is crowdsourcing? [1 mark]

- A. Many residents submit observations through an online platform. B. One computer repeats a calculation.
C. A programmer renames a local variable. D. Two offline processors divide arithmetic tasks.

Answer: _____

4. Which fictional project best illustrates citizen science? [1 mark]

- A. A store changes its logo. B. A processor divides a file into packets.
C. Residents collect shade measurements for an online scientific study. D. A developer tests only personal preferences.

Answer: _____

5. Select exactly TWO justified statements. [1 mark]

- A. Deleting a region label always eliminates bias. B. Comparing outcomes across groups can reveal a gap.
C. A large crowd guarantees true reports. D. Improving coverage can reduce some bias without removing all bias.

Answer: _____

6. All listed access barriers are removed in a fictional model. Why might collected reports still be unrepresentative? [1 mark]

- A. All enabled people must submit. B. A valid file format proves representation.
C. People may differ in willingness or time to contribute. D. Equal access forces identical opinions.

Answer: _____

AP CSP | Lesson 28: Bias & Crowdsourcing

Written responses | Questions 7-9

7. [4 marks]

In a fictional map, only app users can report and ranking uses raw report counts. Identify one data bias risk and one design bias risk, then give a matched action for each.

8. [4 marks]

Synthetic report rates: North 20%, South 80%. Sample A has 80 North and 20 South; B has 40 North and 60 South. Calculate each total report rate, give the difference in percentage points, and explain the cause.

9. [4 marks]

Synthetic population shares are North 60%, South 40%; report rates are 20%, 80%. Calculate the weighted rate, state the condition that keeps it fixed when the sample mix changes, and name one bias it does not remove.

AP CSP | Lesson 28: Bias & Crowdsourcing

Written responses | Questions 10-12

10. [4 marks]

Design a fictional online citizen-science shade study. State a research question, what distributed participants contribute using devices, one benefit of collaboration, and one matched quality check.

11. [4 marks]

A fictional map accepts a report if its date and location fields are complete. Explain what this check can detect, what it cannot prove, one stronger verification step, and one remaining limitation.

12. [4 marks]

Evaluate: "Crowdsourcing removes bias; our weighted 44% proves the real need." The 44% comes from synthetic group shares and fixed report rates. Give two corrections, an inclusion action and a measure to evaluate that action.
